Flood Extent Mapping in Nairobi

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Flood Footprint - Nairobi City

Nairobi, a city of about **5.8 million** inhabitants, is one of Africa's fastest-growing cities.

Torrential rainfall led to flooding, reaching peak levels on **7th March, 2026**.

45 lives were lost, alongside significant destruction of property and infrastructure.

Research Motivation

Gap 1: Assessment Uncertainty

Flood disasters pose significant challenges for urban planning. Existing observations often fail to accurately quantify the spatial footprint of flooding events, especially in dense urban areas.

Gap 2: Lack of Mapping

Many regions, especially rapidly urbanizing areas, lack timely and reliable flood risk mapping.

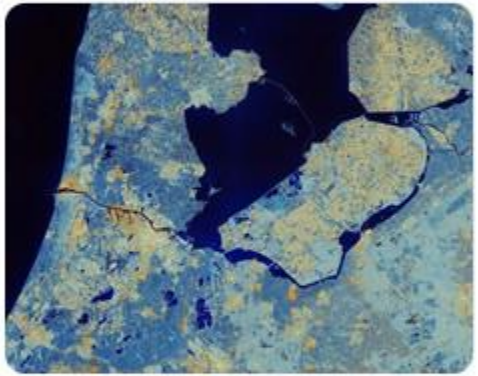
Our Proposed Approach

Explore the integrated use of:

Satellite Imagery + Feature Engineering + Machine Learning + Deep Learning

Aim: To create flood footprints that can aid flood disaster planning efforts.

Data Sources

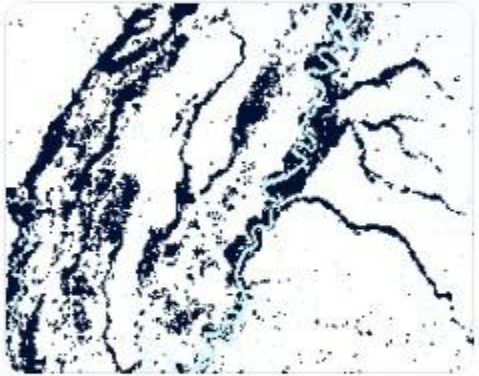


SAR | Sentinel-1

Weather-independent radar imaging. Penetrates cloud cover to detect surface water backscatter.

Sentinel-2

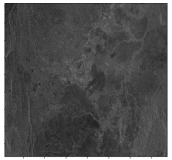
High-resolution optical data. Critical for spectral validation and vegetation monitoring.



Google Research

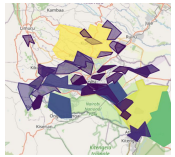
Global Floods Data providing historical benchmarks and event verification datasets.

DATA SOURCES



Sentinel 1

Sentinel 2



Google Flood

UnoStat Nairobi Flood Map

Training

Sen1Flood11

Application

Nairobi

"Ground Truth Data"

MODELING

STAGE 1: THRESHOLD-BASED (Traditional Methods)

Sentinel-1 Difference Thresholding

VV Before – VV After

VH Before – VH After

Sentinel-2 MNDWI Thresholding

$$\text{MNDWI} = \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}}$$

STAGE 2: MACHINE LEARNING (Random Forest)

Sentinel-1

- VV
- VV / VH
- VH
- NPD
- VV -VH

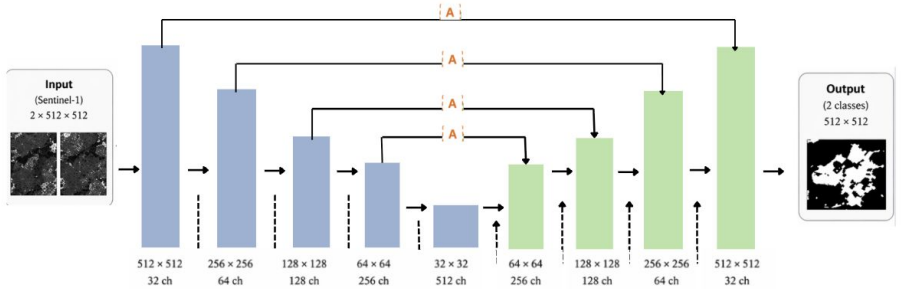
Sentinel-2

- 13 raw bands
- NDWI
- MNDWI
- NDVI
- AWEI

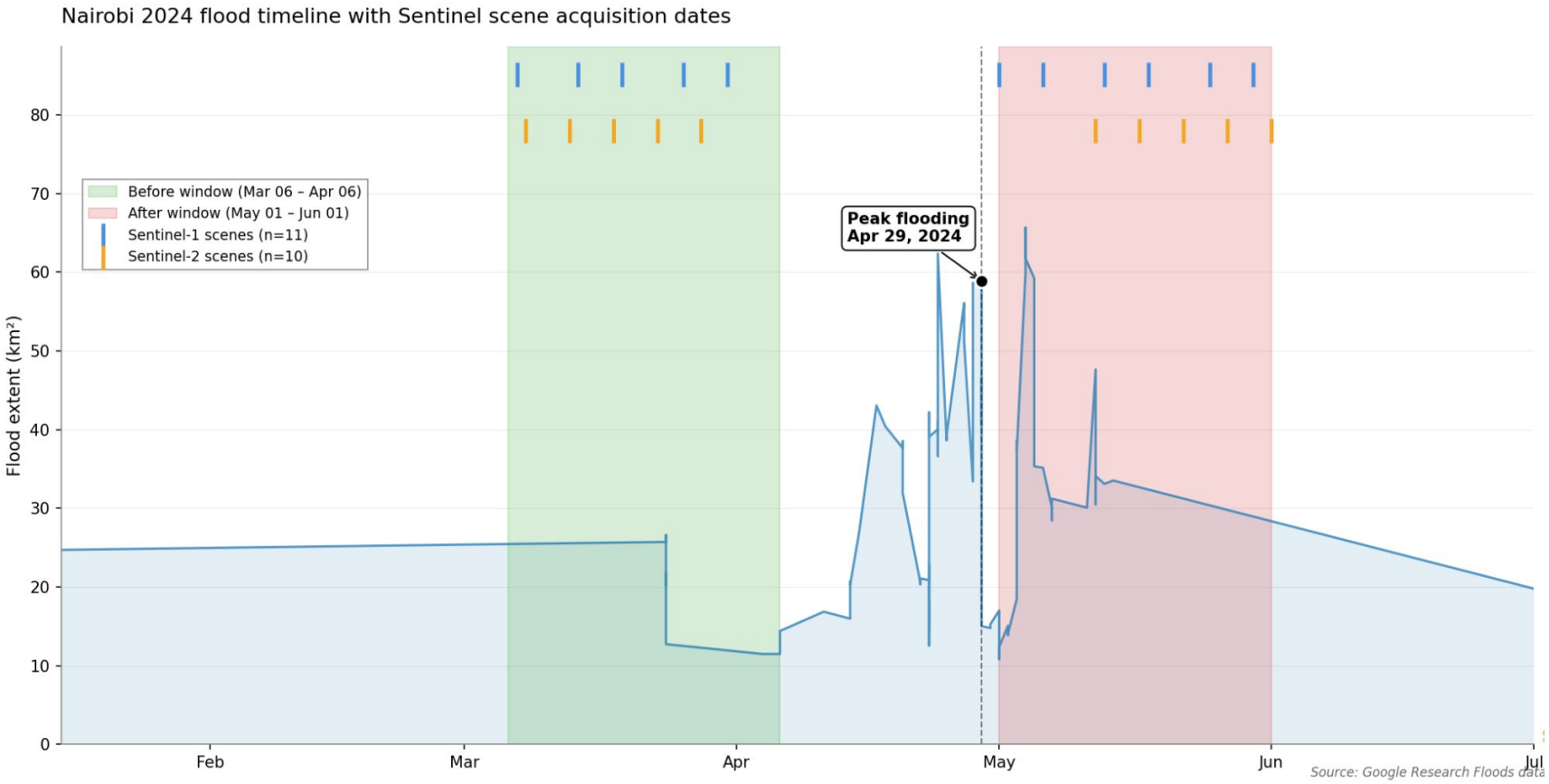
Fusion

- All 22 Features

STAGE 3: DEEP LEARNING



- Residual Block (Encoder)
- Attention Up Block (Decoder)
- Max Pooling 2 × 2
- Transpose Conv 2 × 2
- Attention Gate
- Conv 3 × 3, BN, ReLU



Source: Google Research Open Buildings & flood extent dataset

Threshold

STAGE 1: THRESHOLD-BASED (Traditional Methods)

Sentinel-1 Difference Thresholding

VV Before – VV After

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Sentinel-2 MNDWI Thresholding

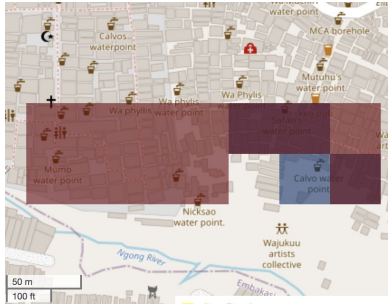
$$\text{MNDWI} = \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}}$$

Comparing Indices

VH:	3.41 km²	(	3,055 px)
VV:	0.80 km²	(	715 px)
MNDWI flood:	9.30 km²	(	8,343 px)

	VH	VV	MNDWI
VH	1.00	0.01	0.11
VV	0.01	1.00	0.01
MNDWI	0.11	0.01	1.00

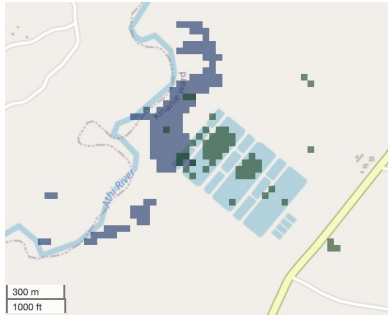
Mismatch Examples



MNDWI

VV

VH

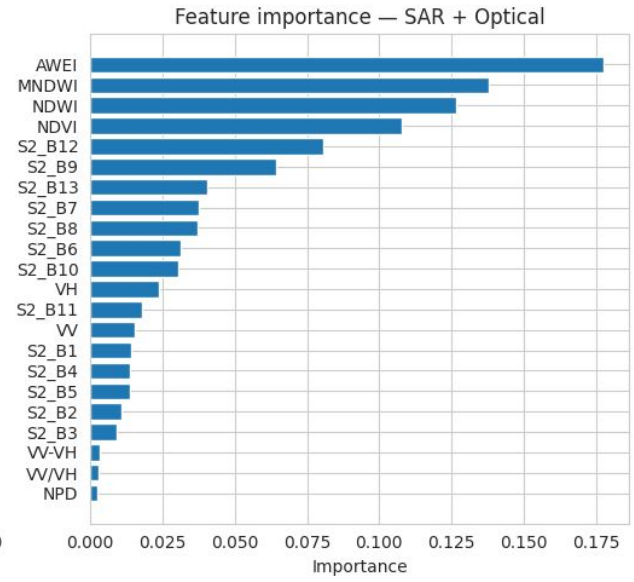


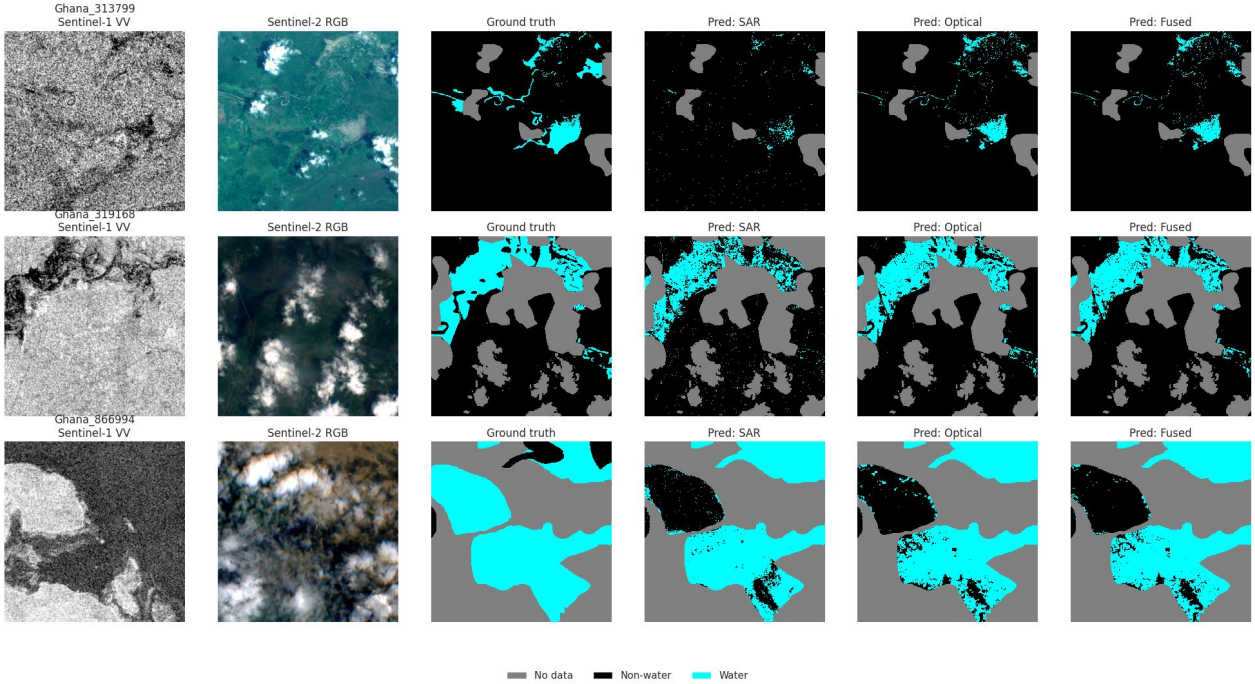
Machine Learning

Training Result

STAGE 2: MACHINE LEARNING (Random Forest)

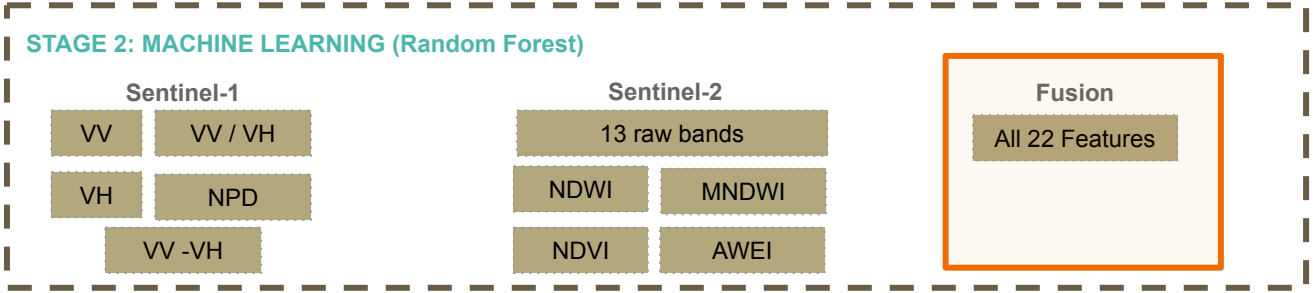
Sentinel-1	Sentinel-2	Fusion
VV	13 raw bands	All 22 Features
VV / VH	NDWI	
VH	MNDWI	
NPD	NDVI	
VV -VH	AWEI	





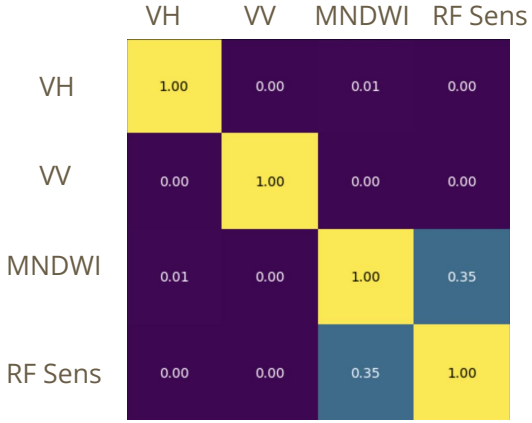
Machine Learning

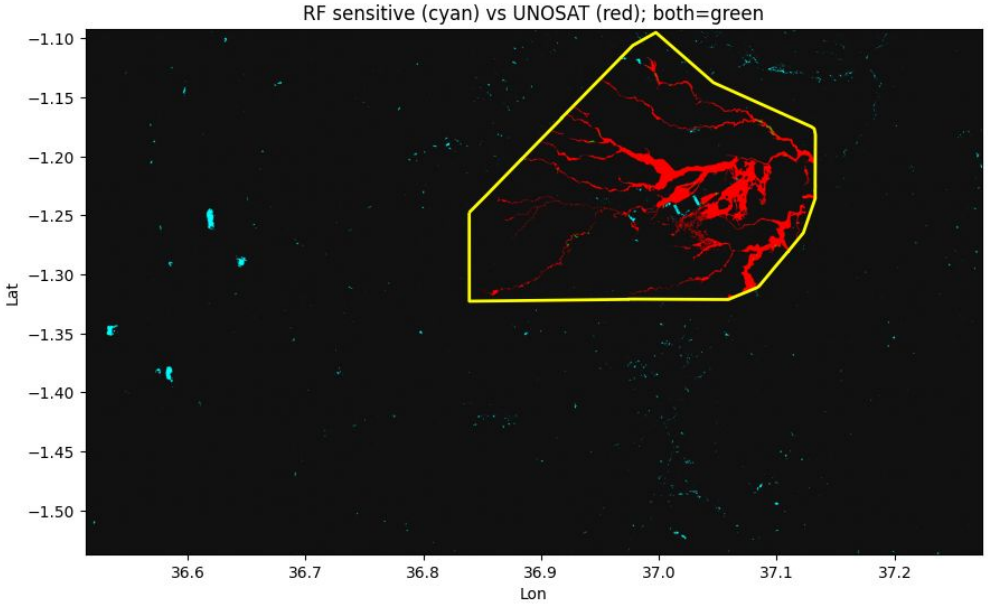
Prediction Result



Comparing Random Forest Results

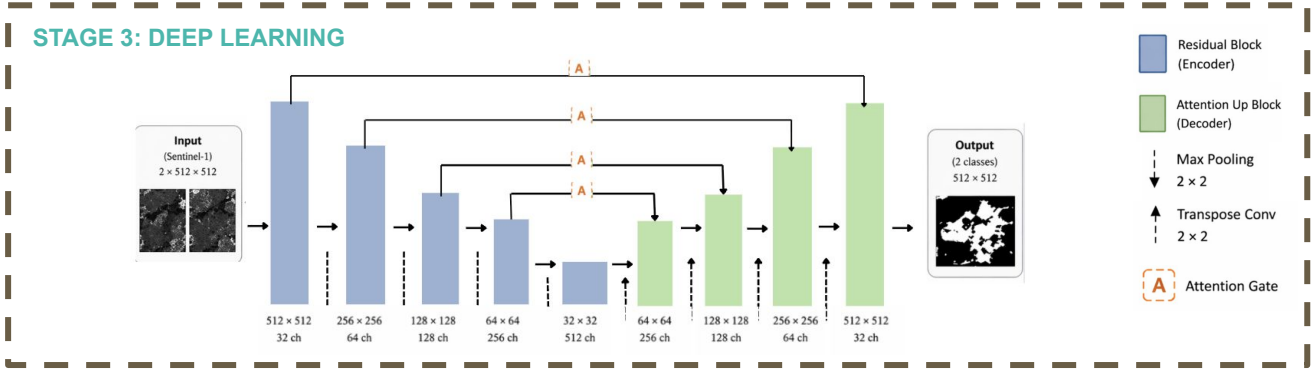
RF conservative	9.86 km²	(8,846 px)
RF sensitive	10.01 km²	(8,979 px)
RF agree	9.52 km²	(8,536 px)





Deep Learning

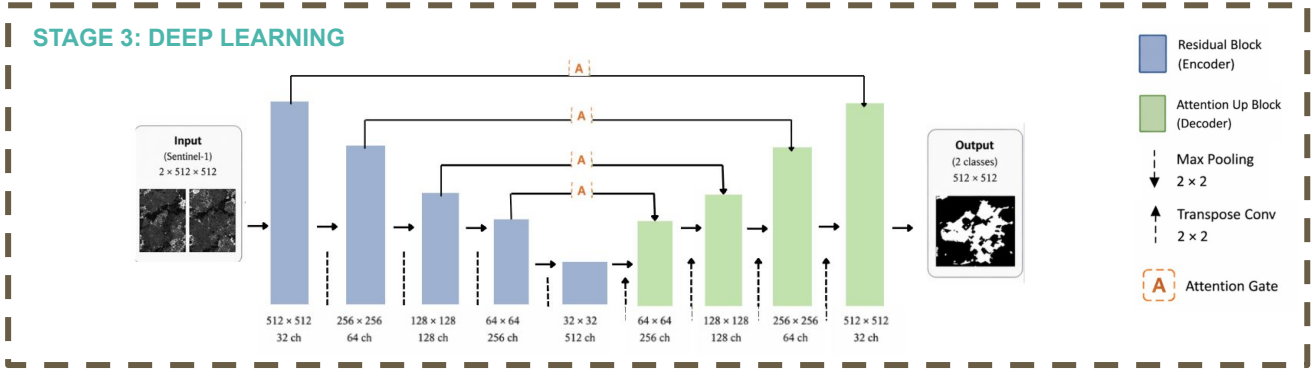
Training Result

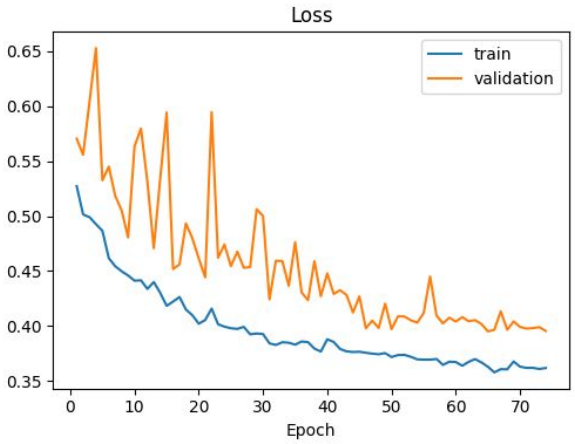


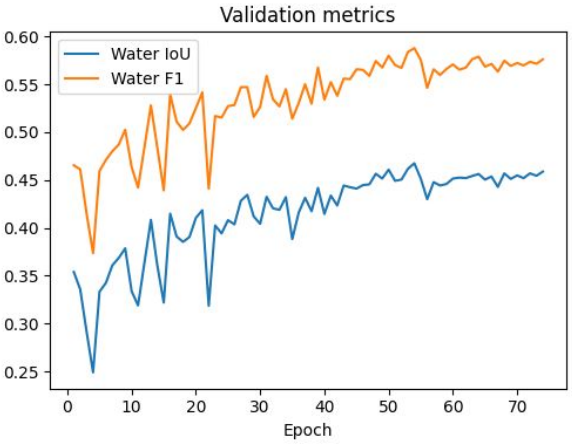
Rank	Experiment	Model	Loss	Best threshold	IoU	F1	Precision	Recall	Pred water frac
1	Exp 6	Residual Attention U-Net	Weighted CE + Dice 0.25	0.45	0.548	0.708	0.724	0.693	0.106
2	Exp 2	Small U-Net	Manual WCE [1,3]	0.55	0.536	0.698	0.757	0.648	0.094
3	Exp 1	Small U-Net	Auto WCE	0.45	0.528	0.691	0.722	0.662	0.101
4	Exp 5	Residual Attention U-Net	Auto WCE	0.55	0.521	0.685	0.737	0.639	0.096
5	Exp 4	Small U-Net	Auto WCE + Dice 0.25	0.55	0.513	0.678	0.724	0.637	0.097
6	Exp 3	Small U-Net	Auto WCE, LR=5e-4	0.60	0.491	0.659	0.773	0.573	0.082

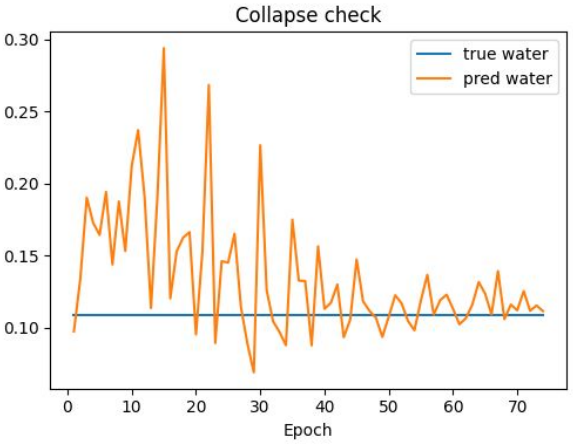
Deep Learning

Training Result





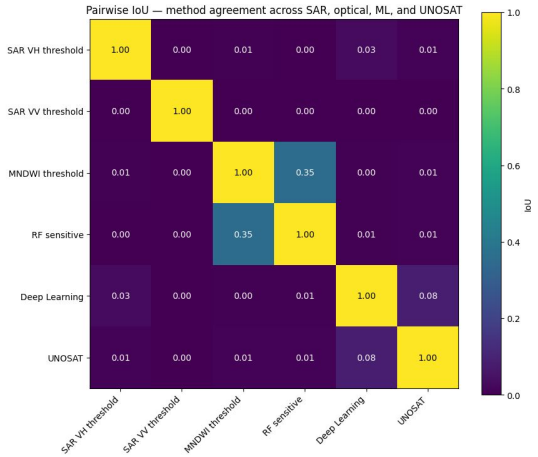




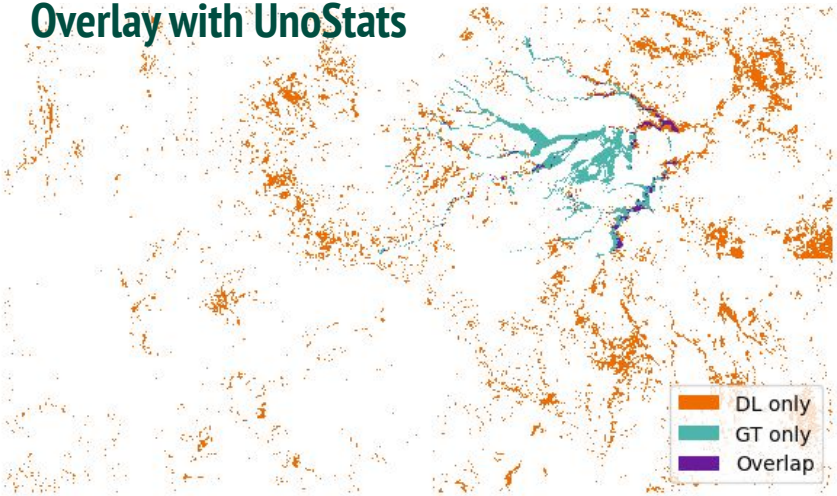
Deep Learning

Prediction Result

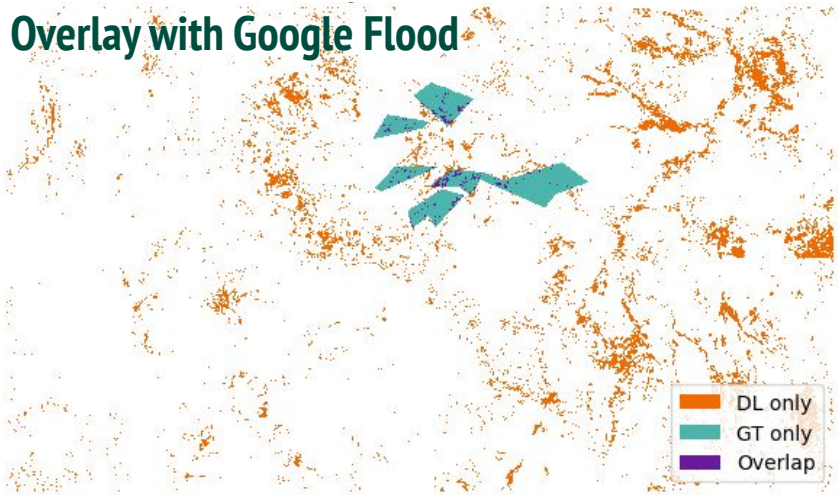
SAR VH threshold	3.41 km²	(3,055 px)
SAR VV threshold	0.80 km²	(715 px)
MNDWI threshold	9.30 km²	(8,343 px)
RF agree	9.52 km²	(8,536 px)
Deep Learning	**251.39 km²**	**(225,466 px)**



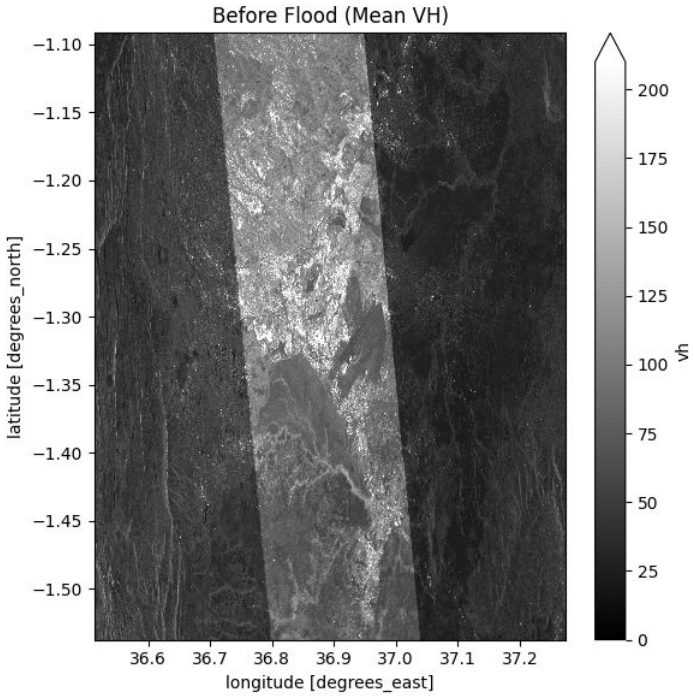
Overlay with UnoStats

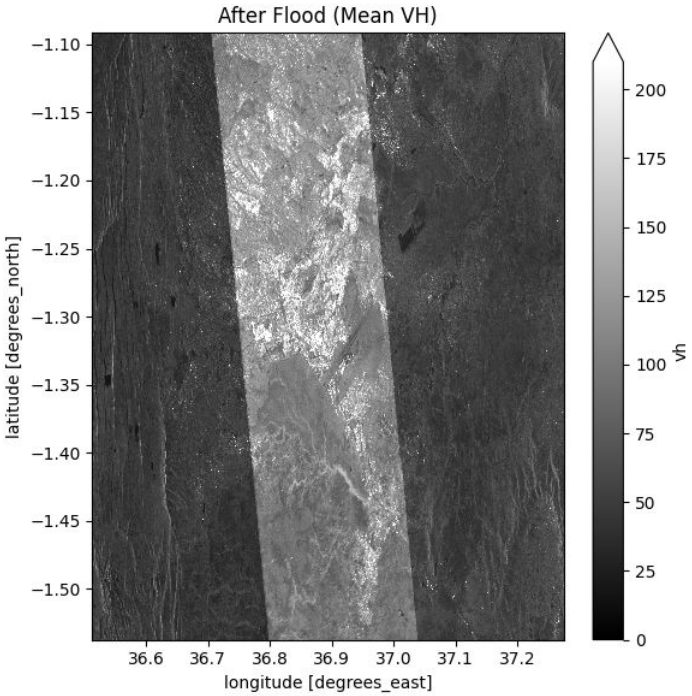


Overlay with Google Flood



discuss limitations honestly





Limitations & Challenges

Time Window Trade-offs

The "before" window straddled the onset of long rains, making the baseline partially wet.

Challenges: Smaller windows limit scene counts, while larger windows risk seasonally inappropriate water levels.

Ground Truth Mismatch

UNOSAT (50cm Pléiades):
Based on a single event, whereas our model estimates general seasonal flood extent.

Google Flood Reports:
Coarse polygons aggregated over large events rather than pixel-level delineations.

Training Data Bias

Significant mismatch between training data and Area of Interest (AOI).

Sen1Floods11 skews rural-tropical, resulting in limited model exposure to "urban confusers."

Takeaways